

MODEL QUESTION PAPER

REVISION -2010  
SUB CODE -3080

Reg. No. : .....  
Signature: .....

THIRD SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

**FLUID MECHANICS**

TIME : 3 Hour

MAXIMUM MARKS :100

**PART-A**

I (Answer all questions . Each question carries 2 marks) (Marks)

1. Define density.
2. Define critical velocity
3. Write Bernoulli's equation
4. Define skin friction
5. Write Fanning's Equation

**PART-B**

II (Answer any five of the following questions, each question carries 6 marks)

1

- a) Compare Newtonian and non Newtonian fluids
- b) Density of water at  $4^{\circ}\text{C}$  in MKS system is  $1000\text{ kg/m}^3$ . Calculate the specific gravity of chloroform. Density of chloroform is  $1.489\text{ kg/m}^3$  at  $20^{\circ}\text{C}$ .

2

- a) Explain U-tube manometer
- b) Define total pressure

3

- a) Compare viscous flow and turbulent flow
- b) Define Reynolds' number

4

- a) Define Equivalent length
- b) State expansion losses

5

- a) Describe V – notch
- b) Draw the sketch of Rota-meter

6

- a) State impact head.
- b) Define Net positive suction Head

7

- a) Derive Bernoulli's equation.

### **PART-C**

III

(Answer one full question from each module. Each question carries 15 marks)

#### **MODULE I**

- 1. Draw a neat sketch and explain Redwood viscometer. (8)
- 2. Discuss velocity distribution in pipeline (7)

**OR**

IV

- 1. Calculate pressure difference using inclined tube manometer (6)
- 2. A fluid of density  $0.9 \text{ gm/cc}$  and viscosity  $0.1 \text{ poise}$  is flowing through a pipe of diameter of  $5 \text{ cm}$  at the rate of  $1 \text{ L/minute}$  whether the flow is viscous or turbulent. (6)
- 3. Define gauge pressure (3)

#### **MODULE-II**

V

- 1. Calculate friction losses in straight pipes using Fanning's equation (6)
- 2. List the classification of flow measuring equipments (3)

3. Explain the methods to calculate losses due to expansion in pipes (6)

**OR**

VI

1. Explain the working of a current meter (8)
2. Draw a neat sketch and explain rectangular notch (7)

**MODULE-III**

VII

1. Draw a neat sketch and explain butterfly valve (8)
2. Explain the constructional details and working of internal gear pump (7)

**OR**

VIII

- Write short note on (1) Air lift pump (2) Acid egg pump (3) lobe pump (15)

**Module IV**

IX

1. Explain the working of reciprocating compressor with fig (8)
2. Explain the constructional details and working of centrifugal blower (7)

**OR**

X

1. Explain the working of single stage ejector, with neat sketch (8)
2. Explain the working of centrifugal fan with fig. (7)